

Learning Under Hardware Shift: Neural Decoding Across Superconducting Quantum Processors

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Abstract

We study supervised learning under temporal and cross-device distribution shift using quantum-error-correction syndromes as structured spatio-temporal data. The practical task is to reduce logical error rates, which must approach 10^{-12} per operation for fault-tolerant applications, far below today's 10^{-3} to 10^{-2} physical error rates. The standard decoder, minimum-weight perfect matching (MWPM), uses a fixed noise model that cannot capture every correlated error on real hardware. Prior neural decoders (convolutional, recurrent, graph-structured, transformer) have each improved on MWPM in simulation. To our knowledge, no prior work combines all three elements tested here: direct comparison of multiple machine-learning (ML) decoders, confirmation on untouched real-hardware blocks, and transfer across physical backends without retuning. This paper addresses that gap. Our proposed residual decoder encodes each syndrome as numerical (x, y, t) detector tokens and processes them with three graph-biased transformer blocks (six attention heads), learning when to correct MWPM's output. A label-free drift guard and two XGBoost benefit selectors route uncertain cases to analytical experts. We freeze every learned component before testing on independent blocks from three backends, on its training backend, the neural decoder repeatedly cuts logical error below MWPM by 1.34 percentage points (4.0% relative) across 259,200 shots. Transferred without retuning, its benefit shrinks on one backend and disappears on another, while backend-specific classical experts recover the gain where the learned policy fails. Further architectural and quantum-assisted extensions did not improve on this result and are left for future work. The results identify the best-performing confirmed ML decoder here and show its benefit is conditional on hardware drift from the training distribution, motivating domain-aware model selection over one universally deployed model.

Keywords: machine learning; distribution shift; transformer; quantum error correction (QEC); surface-code decoding; logical error rate; minimum-weight perfect matching (MWPM)

1. Introduction

This paper studies a machine-learning problem with broad deployment relevance: supervised classification when the data-generating system drifts over time and differs across devices. Quantum error correction (QEC) provides a demanding testbed in which structured spatio-temporal parity data must be decoded under calibration drift and hardware heterogeneity. We ask whether learned neural decoders outperform fixed analytical models and, critically, whether any gain survives on execution blocks and physical processors excluded from training and model selection.

Reducing decoding error is central to fault-tolerant quantum computation: quantum information is stored redundantly, but that protection works only when a decoder correctly infers errors from measured parity checks. Demanding applications require logical error rates near 10^{-12} per operation [1], far below today's 10^{-3} - 10^{-2} physical error rates. Existing machine-learning studies mainly report simulation gains. The unresolved ML question is which model improves on a strong analytical baseline on real hardware and transfers to unseen devices. We therefore train and freeze competing decoders, compare them on real International Business Machines (IBM) hardware, quantify the best gain, and report positive and negative outcomes.

Quantum error correction spreads one logical bit across noisy physical qubits and repeatedly measures binary parity checks. Their sequence is a syndrome history; a change between rounds is a detection event [2]. For ML, this history is the structured input and the required correction is the binary target. Surface-code distance d measures redundancy and robustness [3]. Minimum-weight

perfect matching (MWPM) [4, 5], the standard analytical baseline, pairs detection events on a spacetime graph using fixed noise-model weights; learned decoders instead estimate hardware-specific correlations from data.

Framed as a learning problem, the decoder must generalize from a labelled training distribution to test-time inputs drawn from a shifting hardware distribution: superconducting qubits suffer from gate error, measurement error, decoherence, leakage, crosstalk, and calibration drift, all of which perturb the input distribution the model was trained on. This shift risk is present throughout the neural-QEC-decoder literature that motivates this paper's design choices. Convolutional networks applied to local spatial syndrome grids were among the first neural decoders to improve on phenomenological MWPM in simulation [6]. Recurrent architectures then extended this to full multi-round syndrome histories [7], and later work targeted decoders that scale to larger code distances [8]. Building on this lineage, more recent work combines graph-structured detector representations [9, 10], long-range attention [11], and calibration-conditioned feature modulation [12]. AlphaQubit [1] closely approaches a decoder with full knowledge of the noise model at distance 3-5 on Sycamore hardware, using soft analog readout and leakage labels unavailable here.

A central, underappreciated risk in this literature is that a neural gain measured on one hardware block does not necessarily hold on the next block, and even less reliably on a different physical backend. Calibration drift and processor changes can leave a model systematically miscalibrated elsewhere. This paper reports a multi-stage, pre-registered programme addressing this risk directly. We construct an exact-circuit MWPM baseline from the submitted transpiled circuits and calibration snapshot for each job, then evaluate, in strict chronological freeze-then-submit order:

1. A custom multi-head self-attention transformer-style decoder trained as a supervised binary classifier across multiple calibration domains;
2. A frozen cross-domain selector combining that decoder with a gradient-boosted-tree (XGBoost) benefit classifier routing to a classical complementary solver; and
3. The transfer of both frozen Kingston policies to two additional IBM backends without retuning.

Every reported number distinguishes independently confirmed post-freeze evaluation from development or postselected results. Every negative or null outcome is retained and reported, including a small-scale quantum-assisted classifier/Quantum Approximate Optimization Algorithm (QAOA) screen (Section 5.1.3) that did not clear the pre-registered confirmation bar for improving the artificial-intelligence (AI) decoder.

The central question, stated as a machine-learning problem, is whether a neural or hybrid residual classifier, frozen before evaluation, can produce a reproducible logical-error improvement over an exact-circuit MWPM baseline under distribution shift, first across independent execution blocks on one device and then across different physical backends. Section 4.5.3 and Table 6 give the nuanced answer in full; in short, the transformer-style decoder generalizes within-backend but not uniformly across backends, motivating hardware-domain-aware complementarity over one decoder applied everywhere.

ML contributions: (1) we propose a graph-biased residual transformer that models all detector pairs across space and time and learns corrections to an exact-circuit MWPM baseline; (2) we combine it with a label-free drift guard and frozen XGBoost selectors for risk-controlled deployment; and (3) we provide a paired comparison of neural, tree-based, recurrent, convolutional, classical, and quantum-assisted alternatives on untouched hardware blocks. The proposed decoder lowers pooled logical error by 1.34 percentage points (4.0% relative) across 259,200 untouched Kingston shots and by 1.53 points on a fourth block. Cross-backend tests then identify where the learned policy transfers and where domain-specific experts are required. A stronger universal cross-device model is therefore future work, not a claim of this paper.

2. Related Work

2.1 Surface-Code Decoding and MWPM

Minimum-weight perfect matching uses noise-model likelihoods as graph-edge weights [4, 5]. Our exact-circuit baseline derives these weights from each submitted circuit and calibration snapshot. Perturbed baseline members also provide consensus and disagreement features to the learned router, related to matching harmonisation [13].

2.2 Neural and Hybrid Syndrome Decoders

Data-driven decoders treat syndrome histories as structured inputs [9, 10]. Graph models encode spacetime coordinates; transformers [11] model long-range dependencies; and calibration conditioning [12] represents hardware context. AlphaQubit [1] additionally uses analog readout and leakage labels unavailable here. Heavy-hexagonal classification [14] and neural three-dimensional (3D) pre-decoding [15] report simulated gains but not real-hardware cross-processor transfer. These studies motivate our residual model and a graphics processing unit (GPU)-accelerated belief-propagation/ordered-statistics decoder (BP-OSD) as its non-neural complementary expert.

2.3 Calibration-Conditioned Decoding and Distribution Shift

Calibration-conditioned decoding [12] addresses hardware variability via slowly updating hardware context; Feature-wise Linear Modulation [16] is a general conditioning mechanism, here using label-free detector-rate summaries as a hardware fingerprint without accessing logical labels. Distribution shift is an active concern [13, 17]: a model trained on one calibration block may overfit correlations specific to that environment, and random within-block splits mask this risk, motivating both the label-free shift guard and the cross-backend generalization test central to this paper.

2.4 Selective Prediction and Risk-Controlled Deployment

Selective prediction [18] defers a classifier's output when the input distribution shifts outside its validated domain. Applied here, selective decoding invokes a learned policy only when a label-free context test confirms the current hardware distribution is close enough to the training domain, falling back to a strong analytical decoder otherwise, used both as a per-cohort shift guard and, more strictly, as a label-free high-reliability postselection mode.

3. Proposed ML Framework and Evaluation Design

Figure 1 summarizes the proposed ML pipeline and comparison design. One shot is one circuit execution and therefore one labelled example; a job or block is a batch collected from one backend (physical processor) under one calibration snapshot. Each shot becomes detector tokens (x, y, t) , where t indexes repeated syndrome-measurement rounds. Exact-circuit MWPM supplies the baseline and residual target; the relational transformer predicts when it should be corrected, while a label-free drift guard and XGBoost selectors route unsafe cases to MWPM or the belief-propagation/ordered-statistics decoder (BP-OSD). All components are frozen before paired tests. Kingston supplies training/validation domains; later Kingston jobs plus Fez and Marrakesh are untouched test domains. The six cohorts combine X- or Z-basis memory with 3, 5, or 7 syndrome rounds. All tests use code distance 3; distance 5 is excluded because routing overhead drove failure close to chance.

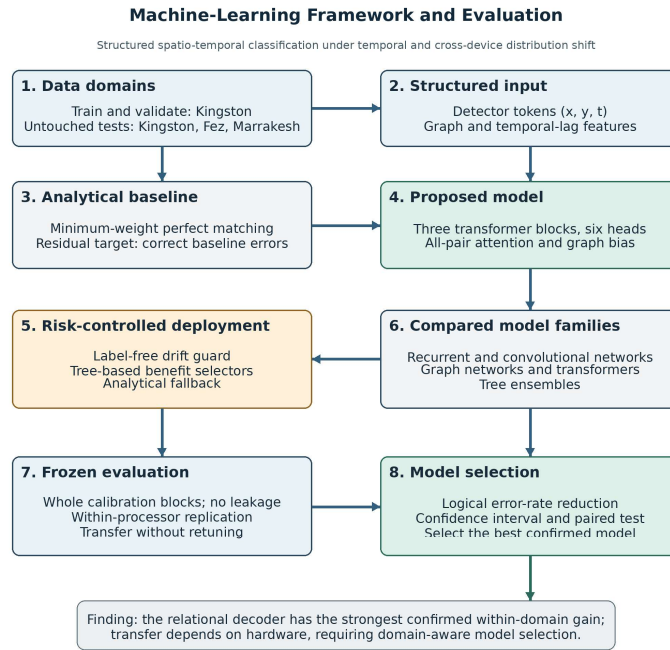


Fig. 1. Proposed ML framework: structured detector tokens, residual transformer, risk-controlled routing, competing model families, and frozen within-domain/cross-device evaluation.

Table 1 summarises the primary confirmed jobs. Every job used 108,000 total shots per block, split into six basis/round cohorts with the first 20% (21,600 shots) reserved for label-free context estimation only and the remaining 80% (86,400 shots) as the untouched scored set, examined only after the relevant policy was frozen. Two additional historical Kingston precursor jobs (Section 4.1) used 108,000-120,000 shots each under the same split discipline.

Table 1. Primary independently confirmed IBM jobs (108,000 shots each: 21,600 context / 86,400 scored).

Job	Processor (backend)	Role
K1	ibm_kingston	relational decoder first confirmation
K2	ibm_kingston	relational decoder / extended relational decoder second confirmation
K3	ibm_kingston	extended relational decoder + local-hypergraph-solver overlay confirmation
K4	ibm_kingston	extended relational decoder + cluster/conformal-hurdle overlay (null) confirmation
K5	ibm_kingston	extended relational decoder + high-reliability postselection confirmation
K6	ibm_kingston	cross-domain selector (extended relational decoder + BP-OSD selector) confirmation
K7	ibm_kingston	extended relational decoder fourth independent block +

		hypergraph-benefit router (null) confirmation
F1	ibm_fez	Hard-syndrome residual expert (Fez-specific) confirmation
F2	ibm_fez	BP-OSD / extended relational decoder / cross-domain selector cross-backend confirmation
M1	ibm_marrakesh	extended relational decoder cross-backend confirmation

Random shot-level splitting within a job was rejected because it leaks calibration-block characteristics across the split; development used whole execution blocks, separated from confirmation by calibration time or backend identity. Every policy in Sections 3-4 was frozen (in standard machine-learning terms, locked prior to test-set evaluation) before its confirmation job was submitted.

3.1 Domain-Level Data Partitioning

No single global train/validation/test percentage applies across this paper, because the split is defined by whole calibration-day hardware domain, not by a shot-level percentage: a percentage split would mix shots from the same calibration domain across the train and validation sets, leaking calibration-specific artifacts and overstating generalization. Instead, the decoders of Section 3.4 are selected by leave-one-domain-out cross-validation over real-hardware calibration blocks, and evaluated exactly once each on an independent test set of calibration blocks the model never saw during training or model selection.

The training and validation sets are built from `ibm_kingston` calibration domains under leave-one-domain-out folds (relational decoder: Model A/B over domains 0-4; extended relational decoder: checkpoints `allpair_holdout3/4/5` over domains 0-5), with per-domain roles tabulated in Table 2. Model selection, early stopping, and threshold calibration in Sections 3.4-3.5 use validation-domain performance only.

The test set is the set of confirmation jobs named in Table 1: K1-K7, F1, F2, and M1. Every one of these jobs is an independently collected IBM execution block that no training or validation step in Sections 3.4-3.5 ever accessed; each is evaluated exactly once under its pre-registered, frozen decoder policy, and every outcome, positive, negative, or null, is reported in Sections 4-5. This matches the standard AI convention of a held-out test set used only for final reported evaluation, not for model selection.

The 20%-shot context prefix described in Section 3.5.1 is a separate, orthogonal, label-free drift-context window collected inside each confirmation job; it supplies unlabeled hardware-drift context to the label-free guard and is never scored, so it is not a fourth data split. Table 2 summarizes the resulting role of every domain and job used in this paper.

Table 2. Data partitioning: training/validation domains and test-set confirmation jobs.

Domain / Job	Role
Domain 0 (<code>ibm_kingston</code>)	Training (relational decoder, extended relational decoder)
Domain 1 (<code>ibm_kingston</code>)	Training (relational decoder, extended relational decoder)
Domain 2 (<code>ibm_kingston</code>)	Training (relational decoder, extended relational decoder)
Domain 3 (<code>ibm_kingston</code>)	Training (relational decoder Model A); validation (relational decoder Model B; extended relational decoder <code>allpair_holdout3</code>)
Domain 4 (<code>ibm_kingston</code>)	Validation (relational decoder Model A); training (relational decoder Model B); validation (extended relational decoder <code>allpair_holdout4</code>)

Domain 5 (ibm_kingston)	Training only (extended relational decoder; reused former relational decoder confirmation block)
K1-K7	Test (confirmation jobs, ibm_kingston, evaluated once each)
F1, F2	Test (confirmation jobs, ibm_fez, evaluated once each)
M1	Test (confirmation job, ibm_marrakesh, evaluated once)

3.2 Analytical Baseline and Residual Target

This section constructs the non-learned baseline against which the trained decoder of Section 3.4 is measured throughout the rest of the paper. For every job, archived QPY circuit files (IBM’s serialized circuit format) were parsed to extract the exact gate/measurement/reset sequence, and the backend calibration snapshot supplied operation-specific error probabilities, mapped to a Stim [19] detector-error model (DEM) and decoded with PyMatching [4, 5]. This exact-circuit approach is more faithful to hardware execution than a phenomenological model, capturing layout-specific error propagation including heavy-hex routing; a new DEM was reconstructed independently for every job, including every cross-backend confirmation.

An auxiliary 33-member exact-circuit MWPM ensemble, built by deterministically perturbing operation-level error probabilities around nominal calibration values, supplied per-member matching-weight statistics as downstream input features; its majority vote differed from nominal MWPM by under 0.02 points on representative blocks, so nominal MWPM, selected on development data before any confirmation job, remains the pre-registered primary baseline throughout.

3.3 Frozen Evaluation Protocol and Statistics

All architecture, threshold, and routing decisions were frozen and hashed on development data before each confirmation job; frozen means no parameter, threshold, or routing rule changed after submission, and untouched means test examples were not examined beforehand. Each circuit encodes a known value and produces parity-check features; the decoder infers a correction without seeing the target bit. A logical error occurs when the corrected output disagrees with the encoded value. Logical error rate (LER) is the fraction of scored shots that are wrong, so lower is better. Gain in percentage points (pp) is Baseline LER minus candidate LER, so positive gain means improvement. Every method is scored on identical shots. A 95% confidence interval (CI) gives uncertainty around the gain; an interval excluding zero together with a paired two-sided McNemar p-value below 0.05 is treated as evidence of improvement. CIs use 10,000 paired bootstrap resamples, and all six cohorts are also checked for regressions.

3.4 Proposed Graph-Biased Relational Decoder

This section describes the neural decoder’s input representation and the frozen architecture built on top of it, the core learned component evaluated throughout the rest of the paper.

3.4.1 Structured Detector-Token Representation

Detection events are encoded as numerical tokens with coordinates (x, y, t): spatial position on the surface-code patch and syndrome-round index. This yields a graph-structured, spatio-temporal input: a detector graph derived from circuit geometry defines connectivity, providing a physically motivated attention bias over nodes located in space (x, y) and time (t). Numerical (x, y, t) coordinates, rather than any rendered image, are the input representation used throughout the final architecture (Section 5.1.1 reports the controlled comparison behind this choice).

3.4.2 Model Construction and Training

The relational decoder (extended in a second iteration as the extended relational decoder) is a multi-head self-attention transformer-style encoder (implemented in PyTorch), the outcome of an iterative programme testing whether modelling long-range, all-pair relationships among syndrome snapshots outperforms both a purely recurrent model and a rendered-image representation (Section 5.1.1). The frozen architecture combines:

- **Explicit all-pair relational attention:** A stack of 3 transformer-encoder blocks (pre-LayerNorm, width 96, 6 attention heads of width 16, 4x-expansion GELU feed-forward sublayer, residual connections throughout, roughly 1.04M parameters), in which detector-snapshot tokens (one token per detector per syndrome round) attend to all other snapshots in the syndrome history via standard scaled dot-product multi-head self-attention, augmented with a learned additive bias keyed on temporal lag and explicit pairwise detector coactivity/change features rather than a conventional positional encoding, a form of sequential/temporal dynamic-data modelling over the syndrome history. This was the single largest driver of development gain across the early relational-ensemble progression that preceded the relational decoder.
- **Detector-geometry graph bias:** Attention is biased using the circuit-derived detector graph of Section 3.4.1: numerical (x, y, t) detector coordinates and multi-lag coactivity maps are fed into the token embedding and attention bias rather than processed by a separate graph-neural-network layer.
- **Multi-day / multi-domain training:** The relational decoder was trained across multiple calibration domains under a leave-one-day-out protocol, addressing the transfer failure of earlier single-domain models (Section 4.1).
- **Exact-MWPM fallback for one unstable configuration:** The X-basis, seven-round (X7) cohort was assigned MWPM fallback because it did not reach the reliability bar met by the other five cohorts during development.
- **Extended-decoder extension:** The extended relational decoder retains the relational decoder's architecture and adds an independently trained expanded-calibration-domain family, combined at a frozen 80/20 blend fixed before any confirmation job.

Training is supervised binary classification: each model head predicts a residual logical flip (1 when MWPM disagrees with the true label, 0 otherwise), optimised with a weighted sum of binary-cross-entropy losses on residual, direct, and risk prediction heads via AdamW (learning rate $3e-4$, weight decay $3e-3$, gradient-norm clipping, mixed-precision training), with early stopping on validation loss. Inference applies a fixed decision threshold of 0.5075 (both the relational decoder and the extended relational decoder) fixed during development. Model, threshold, and cohort-fallback assignment were frozen before each confirmation job below.

3.5 Risk-Controlled Deployment Under Distribution Shift

Two mechanisms guard against the distribution-shift risk identified in Section 2.3: a label-free shift check that falls back to MWPM when the current hardware distribution drifts too far from the training domain, and a learned selector that routes shots to a classical complementary solver when it is likely to help.

3.5.1 Label-Free Drift Guard

The first 20% of each cohort's shot stream (context prefix) is used only for label-free context estimation; its labels are never accessed. The shift ratio, the ratio of mean per-detector activation rate in the context prefix to the training domain's rate, triggers a fallback to unmodified MWPM when far from 1.0. Frozen acceptance thresholds per cohort were fixed on development data before any confirmation job.

3.5.2 Complementary Analytical Expert (BP-OSD)

BP-OSD is a classical, non-neural baseline decoder used here for comparison against and complementarity with the learned decoder, not a learned component. An open GPU-accelerated sum-product belief propagation plus ordered-statistics decoder at order 10 (BP-OSD) was evaluated as an independent classical expert, with detector-error priors from the same exact-circuit DEM. Six BP variants were screened; only sum-product BP with OSD-10 improved on MWPM and was carried forward.

3.5.3 Frozen XGBoost Benefit Selector

The extended relational decoder and BP-OSD independently disagreed on 23-27% of shots across development blocks, indicating complementary error structure. Two gradient-boosted-tree benefit

selectors (XGBoost [20]), trained in opposite hardware-domain directions and calibrated reciprocally, route a shot from the extended relational decoder to BP-OSD only when both agree and pre-frozen thresholds are met; this combination is the cross-domain selector referenced below.

4. Comparative Results

This section compares within-backend replication, cross-backend transfer, selectors, postselection, and negative controls. Fig. 2 plots the effects; Table 6 (Section 4.6) compares every method.

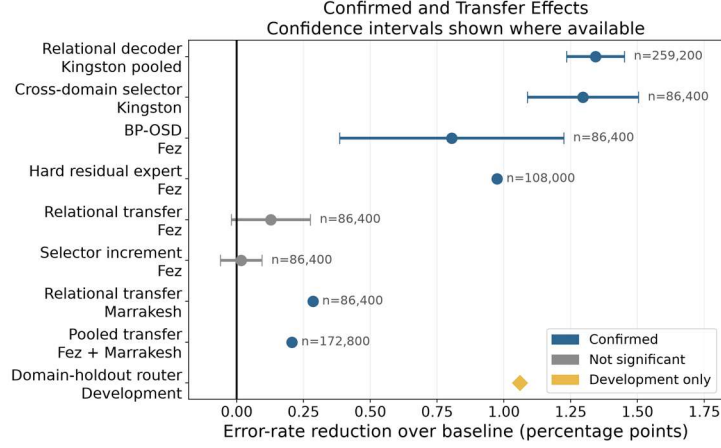


Fig. 2. Error-rate gains over the analytical baseline. Positive values favour the candidate; bars show 95% confidence intervals where available. Diamonds are development-only results and are not pooled with confirmed results.

4.1 Precursor Result and Negative Control

An earlier four-expert safe residual ensemble, guarded by the Section 3.5.1 shift-ratio mechanism, was confirmed on an earlier Kingston job: MWPM 37.91% to 37.26% (+0.65 pp, 95% CI 0.53-0.77; $p < 0.001$, 86,400 shots); the guard enabled the ensemble only for Z7 (41.65% to 37.75%, +3.90 pp). A subsequent frozen explicit-relational ensemble reached the largest pooled *development* gain of the programme (1.81 pp) but *regressed* by 0.65 pp on a fresh Kingston job, motivating the relational decoder's design of Section 3.4.2.

4.2 Pooled Relational-Decoder Confirmation

Frozen relational decoder on K1 improved logical error over MWPM (Table 3); all five enabled cohorts improved, X7 kept its MWPM fallback. K2 improved under both the relational decoder and the extended relational decoder (Table 3).

K3, K4 re-confirmed the extended relational decoder while testing small overlay extensions (Section 5): K3 34.58%→33.39% (+1.19 pp); K4 34.68%→33.15% (+1.53 pp, $p < 0.001$). Pooled K2-K4 (172,800 shots) confirmed a consistent gain (Table 3).

K5 doubled as the Section 4.4 postselection test: 31.78%→30.47% (+1.31 pp, $p < 0.001$). Pooled K3-K5 (259,200 shots) gave the strongest repeated whole-dataset result and the primary within-backend claim (Table 3; also Table 6). K7 reconfirmed independently, reported separately rather than pooled into the headline (Table 3).

Table 3. Confirmed gains on `ibm\_kingston` (training processor). Each job is a separate test block. LER is the wrong-shot fraction; gain is baseline minus model LER in pp, so positive is better. CI and McNemar p quantify paired uncertainty.

Job	Model	Baseline LER	Model LER	Gain (pp)	95% CI	McNemar p	Test shots
K1	Relational decoder	35.00%	33.64%	+1.36	1.17-1.55	p < 0.001	86,400
K2	Relational decoder	36.41%	35.64%	+0.77	0.59-0.94	p < 0.001	86,400
K2	Extended relational decoder	36.41%	35.53%	+0.88	0.70-1.06	p < 0.001	86,400
K2-K4 pooled	Extended relational decoder	34.63%	33.27%	+1.36	1.22-1.49	p < 0.001	172,800
K3-K5 pooled	Extended relational decoder	33.68%	32.34%	+1.34	1.23-1.45	p < 0.001	259,200
K7	Extended relational decoder	32.76%	31.23%	+1.53	1.34-1.72	p < 0.001	86,400

4.3 XGBoost Selector Confirmation

On K6, the frozen cross-domain selector was evaluated alongside its two constituent experts on identical shots (Table 4).

Table 4. Selector components on untouched job K6. LER is the wrong-shot fraction; positive gain means fewer errors than exact-circuit MWPM.

Decoder	LER (lower better)	Gain vs. baseline (pp)
Exact-circuit MWPM	35.72%	--
BP-OSD (fixed priors)	35.00%	+0.71 pp
Frozen extended relational decoder	34.59%	+1.12 pp
Frozen cross-domain selector	34.42%	+1.30 pp

The cross-domain selector's paired 95% CI versus exact MWPM was 1.08692-1.50336 pp ($p = 3.60 \times 10^{-34}$); its increment beyond the extended relational decoder alone was 0.17 pp (95% CI 0.08-0.27; $p < 0.001$). The selector routed 1.97% of shots (1,705/86,400), correcting 927 extended-relational-decoder errors against 778 damaged predictions (net 149), strongest on X3 and Z7 (+0.64, +0.36 pp); X7 changed by -0.03 pp and several small-cohort intervals crossed zero.

4.4 Selective-Prediction Result

A separately frozen label-free failure-risk policy, ranking shots using only model confidence and detector-activity features, was frozen at 5% coverage (the retained fraction of shots) and evaluated on K5. Of 86,400 shots, it retained 4,320 with 7 errors: conditional logical error **0.16%** (one-sided 95% upper bound 0.30%). MWPM and the extended relational decoder gave identical predictions on every retained shot, so this identifies a reliably easy subset via label-free features rather than an incremental decoding gain, and is not a whole-dataset rate.

4.5 Cross-Processor Generalization

The central open question is whether policies frozen on `ibm_kingston` retain their effect on a different backend without retuning, and whether a backend-specific expert can succeed where transfer does not.

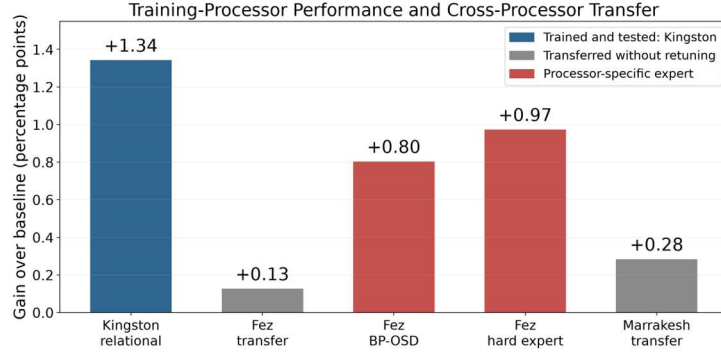


Fig. 3. Kingston-trained extended relational decoder transfers weakly to Fez but retains a small significant effect on Marrakesh; backend-specific experts (BP-OSD, hard-residual) recover a significant gain on Fez where the transferred policy does not.

4.5.1 Fez: Null Learned-Policy Transfer

Two independent Fez jobs isolate where a backend-specific expert succeeds and where the Kingston-trained learned policy does not.

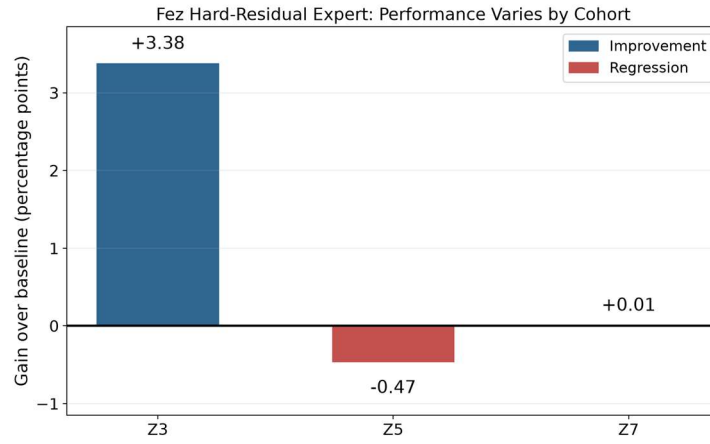


Fig. 4. Per-cohort heterogeneity of the hard-syndrome residual expert on `ibm_fez` (mean of two logical initialisations per basis/round cohort).

On F1, an independently trained gradient-boosted-tree (XGBoost) hard-syndrome residual expert, frozen against Fez syndrome statistics using only hard detector information, reduced MWPM from 46.01% to 45.04% (**+0.97 pp**, 108,000 shots), showing a backend-specific expert can succeed on a second device when trained against its own statistics.

On a second Fez job (F2), the Kingston-trained extended relational decoder policy, the cross-domain selector, and a Fez-specific frozen BP-OSD expert (priors reconstructed as in Section 3.5.2, no retuning) were evaluated simultaneously on the same 86,400 shots (Table 5).

Table 5. Transfer to `ibm\_fez` (a different processor), untouched job F2. Positive pp gain lowers LER; a CI crossing zero or McNemar $p \geq 0.05$ means improvement is not established.

Decoder	LER (lower better)	Gain vs. baseline (pp)	95% CI	McNemar p	Interpretation
Exact-circuit MWPM	46.51%	--	--	--	--
BP-OSD	45.71%	+0.80 pp	0.38-1.22	$p < 0.001$	Independently positive
Frozen extended relational decoder (Kingston-trained)	46.39%	+0.13 pp	-0.02-0.28	$p = 0.10$	Positive estimate, not significant
Frozen cross-domain selector	46.37%	+0.14 pp	--	$p = 0.09$	Positive estimate, not significant
Cross-domain selector vs. extended relational decoder (same shots)	--	+0.02 pp	-0.06-0.09	$p = 0.71$	Cross-backend incremental null

BP-OSD's aggregate positive effect on Fez was configuration-heterogeneous: per-cohort gains ranged from +5.19 pp (X3) down to a regression of -0.99 pp (Z3), ruling out uniform cross-backend improvement and motivating configuration-aware over backend-blanket deployment.

Combined with F1, classical decoder diversity transfers to a new backend better than a routing policy learned on the first backend's error correlations. We treat this cross-backend null as scientifically important as the positive within-backend results, since it isolates the boundary of the method's validity.

4.5.2 Marrakesh: Reduced but Significant Transfer

On M1, the Kingston-trained extended relational decoder policy, applied without retuning, reduced MWPM from 45.54% to 45.25% (**+0.28 pp**, 86,400 shots), smaller than any Kingston confirmation but statistically distinguishable from zero. Pooling Fez and Marrakesh transfer gives 0.21 pp across 172,800 shots, reported only as a transfer-magnitude summary; the two backends are not interchangeable.

4.5.3 Domain-Aware Model Complementarity

Long-range relational neural modelling gives a large, repeated improvement on the backend where it was trained (`ibm_kingston`), with smaller real transfer to `ibm_marrakesh` and unreliable transfer to `ibm_fez`. BP-OSD and backend-specific residual experts, refitted per backend, transfer more reliably but require per-configuration awareness since their aggregate gain can mask cohort-level regressions, favouring a deployment architecture that selects among backend-appropriate experts via a frozen, label-free, hardware-domain-aware gate, rather than one decoder assumed to generalise universally: the central claim of this paper. Cast in general terms, this is a robustness, generalization, and uncertainty-quantification problem: a model's benefit is conditional on how far test-time hardware has drifted from its training distribution, and a reliable deployment must quantify and gate on that shift rather than assume transfer.

4.6 Consolidated Model Comparison

Table 6 consolidates every method's confirmed, development-only, or null result from Sections 4-5, extending the effect sizes plotted in Fig. 2; later sections point back to it rather than restating already-tabulated gains.

Table 6. Model-family comparison. Gain is the absolute LER reduction from MWPM in pp; confirmed means evaluated after freezing on an untouched block, while development-only results require independent confirmation.

Method	Type	Best confirmed gain (pp)	Status
Relational decoder	Transformer encoder	+1.36 (K1)	Confirmed
Extended relational decoder	Transformer encoder, expanded domain	+1.34 (K3-K5 pooled, primary)	Confirmed
Cross-domain selector	XGBoost selector + BP-OSD + extended decoder	+1.30 (K6)	Confirmed
BP-OSD	Belief-propagation + ordered statistics	+0.80 (F2)	Confirmed
Hard-syndrome residual expert	Gradient-boosted trees (XGBoost)	+0.97 (F1)	Confirmed
Local hypergraph subsolver overlay	Rule-based overlay on ext. decoder	+0.05 increment (K3)	Confirmed (small)
Cluster/conformal-hurdle overlay	Conformal-prediction cascade	+0.00 (K4)	Null (confirmed)
Hypergraph-benefit router	Extremely Randomized Trees (ExtraTrees) classifier	+0.02 (K7)	Null (confirmed)
Leave-one-domain-out benefit router	Synthetic-pretrained neural router	mean +1.06 (development)	Development only
Local-3D-convolution residual	3D CNN	+0.00	Null (confirmed)
Extended decoder, cross-backend transfer	Same architecture, no retrain	+0.13 (F2, not significant); +0.28 (M1, significant)	Confirmed transfer-degradation finding

5. Comparative Ablations and Negative Results

Three further overlay mechanisms were developed and frozen as exploratory extensions to the confirmed relational-decoder/extended-relational-decoder; each is reported for completeness, but none is elevated to a primary confirmed claim. Table 6 lists each method's headline gain; this section adds the supporting detail behind those figures.

Local hypergraph subsolver (K3): 95% CI 0.01-0.10 ($p = 0.03$; combined 1.24 pp over MWPM), activating on only 1.25% of shots with conditional gain 4.08 pp, not a whole-dataset effect.

Cluster/conformal hurdle cascade (K4, null): 95% CI -0.03-0.04 ($p = 0.89$), an independent null.

Leave-one-domain-out benefit router (synthetic-pretrained, IBM-fine-tuned neural router; development only): evaluated under leave-one-domain-out across three held-out Kingston domains, gains of 0.92, 1.02, and 1.22 pp at coverages of 8.7%, 12.0%, and 15.0% (all CIs excluded zero) development evidence, not a post-freeze confirmation, and the most promising unconfirmed direction (Section 6.1).

5.1 Controlled Architecture and Routing Ablations

Five independent ablation and negative-result screens probe the boundaries of the confirmed gains above: representation choice, transfer of architectures from adjacent domains, small-scale quantum-assisted alternatives, an exhaustive routing search, and a further frozen hypergraph-benefit router.

5.1.1 Numerical Relations versus Rendered 3D Inputs

A controlled, cross-fitted development screen (identical 192,000-shot pool for all conditions) tested whether preserving explicit numerical relationships among syndrome snapshots outperforms a rendered-image representation [21]: rendered 3D projections improved over MWPM by 1.00 pp, per-

round numerical images by 1.30 pp, and explicit all-to-all lag-aware attention (adopted in the relational decoder and extended relational decoder) by 1.52 pp, ahead of both alternatives (see Fig. 5). A matched-capacity 3D-vs-2D convolutional branch gave +0.50 pp but crossed zero ($p = 0.12$); a 3D-syndrome-radiomics ablation gained 0.02 pp ($p = 0.29$); an adapted local-3D-convolution residual on the extended relational decoder [21] gained 0.00 pp (also in Table 6). These four independent tests consistently show rendered or convolutional 3D representations add no statistically established contribution beyond explicit numerical relational attention.

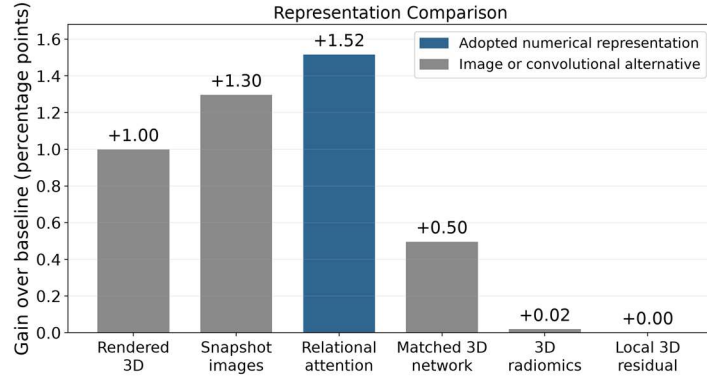


Fig. 5. Rendered and convolutional 3D representations of the syndrome volume did not establish a statistically significant standalone contribution beyond explicit numerical long-range relational attention across four independent tests.

5.1.2 Cross-Architecture Transfer Screen

Five architecture families adapted from other domains, residual 3D convolution [22], a spatial-encoder-plus-bidirectional-recurrent model in the style of convolutional LSTMs [23], an active/high-surprise sparse event model, a relative-spacetime-attention relation-network model [24], and a fusion of all four, were retrained from scratch and tested as candidate additions to the relational decoder under leave-one-domain-out across three held-out Kingston domains. Four of five beat MWPM individually, but every positive family's sign reversed on a held-out domain, and the best blended candidate gave only +0.03 pp (95% CI -0.07 to 0.14, $p = 0.55$): mature architectures from adjacent domains supplied no transferable correction beyond the confirmed relational decoder.

5.1.3 Quantum-Assisted Screen (Negative)

Several small-scale quantum-assisted formulations (classifiers whose parameters run as quantum-gate circuits, plus QAOA, a quantum optimization heuristic, as a sub-solver) were screened against pre-registered classical baselines and did not pass the gate required before committing quantum processing unit (QPU) resources. The classical/AI components alone accounted for every confirmed gain; quantum-assisted augmentation is retained here as a negative result.

5.1.4 Selective-Routing Search

An exhaustive screen of 100 candidate routing policies over the Kingston development pool found none simultaneously positive across every held-out domain and non-regressive in every cohort, reinforcing that configuration- and domain-aware gating (Sections 3.5 and 4.5.3), not a single fixed rule, is necessary.

5.1.5 Hypergraph-Benefit Router (Confirmed Null)

A frozen ExtraTrees benefit router, offering either a minimum-energy or a partition-mass-weighted candidate from a two-hop detector-hypergraph expansion, was confirmed on job K7 alongside the reconfirmation in Section 4.2 (headline gain in Table 6). It widened intervention to 2.01% of shots (1,738/86,400), a null result (95% CI -0.07-0.12; $p = 0.65$). Development coverage on held-out domains (1.86-2.09%) transferred almost exactly to the fresh block, but conditional benefit did not:

positive for X3, near-flat for X7/Z3, negative for X5, showing routing coverage calibrates more easily than counterfactual benefit under hardware drift, consistent with the router nulls of Section 5.

6. Conclusion

We presented a pre-registered machine-learning study of supervised neural and hybrid decoding under temporal and cross-device distribution shift, testing transfer across execution blocks and physical backends rather than reporting one favourable confirmation. The transformer-style decoder of Section 3.4, trained as a supervised binary classifier with an exact-MWPM fallback, produced a repeated gain of 1.34 pp over MWPM across three untouched `ibm_kingston` blocks (259,200 shots). The frozen cross-domain selector combining it with BP-OSD added a further significant 0.17 pp on a fourth block (Table 6). Transferred without retuning, the policy retained a small significant effect on `ibm_marrakesh` but not `ibm_fez`, where a backend-specific BP-OSD expert and an independently trained hard-syndrome residual expert each produced their own gains instead. A representation ablation, a five-family architecture transfer screen, and a small-scale quantum-assisted classifier/Quantum Approximate Optimization Algorithm (QAOA) screen each produced negative or null results reported in full (Table 6): classical gradient-boosted-tree and ExtraTrees selectors improved on the decoder's cross-domain behaviour, while quantum-assisted augmentation did not.

The novelty is the joint evaluation protocol: direct comparison of multiple machine-learning (ML) decoders and a classical baseline on untouched real-hardware blocks, followed by cross-backend transfer without retuning. We found no prior study in the reviewed literature that combines all three elements. Table 6 and Fig. 2 summarize this comparison across every evaluated method. Within it, the extended relational decoder is the best-performing confirmed AI/ML decoder, and a next-generation architecture that closes the remaining `ibm_fez` transfer gap is left as future work (Section 6.1).

From a machine-learning perspective, strong within-domain replication, partial cross-domain transfer, and successful domain-specific alternatives where transfer fails support domain-aware model selection for heterogeneous deployment. The explicit `ibm_fez` null result is therefore as informative as the positive confirmations: it identifies a concrete validity boundary that random within-domain splits would have hidden.

6.1 Limitations and Future Directions

1. **Cross-backend transfer is the central problem:** The Kingston-trained policy transferred with reduced effect to `ibm_marrakesh` and no significant effect to `ibm_fez`; a backend-specific expert was required to recover a gain on Fez.
2. **All primary confirmations use distance-3 circuits;** distance-5 circuits approached 50% logical failure from routing overhead and are excluded.
3. **Absolute error remains far from a sub-1% target:** Aggregate corrected error stays in the 30-35% range; the below-1% postselection result (Section 4.4) is conditional on 5% coverage and is not a whole-dataset rate.
4. **Complete noise characterisation is unavailable:** Pulse-level idle duration, crosstalk tensors, leakage channels, and soft analog readout are not in the exact-circuit model; a prior soft-IQ residual experiment gave only a non-significant gain, and a larger apparent gain was traced to target leakage (a qubit-state artifact correlated with the label, not a genuine decoding improvement) and excluded.
5. **Component ablations are fixed-policy diagnostics,** not causal-contribution estimates under retraining.
6. **No external independent reproduction** has yet occurred.
7. **Latency has not been benchmarked for real-time deployment;** surface-code cycle times are far below current PyTorch inference latency.

Priority future directions: (i) a backend-aware routing gate selecting among Kingston-, Fez-, and Marrakesh-trained experts via label-free backend-identification features, tested on a fourth backend; (ii) a post-freeze confirmation of the leave-one-domain-out benefit router (Section 5); (iii) correlated BP-OSD priors from label-free context shots; (iv) leakage-safe analog readout integration.

Generative AI Use Disclosure

Large language model tools were used during the preparation of this manuscript to assist with literature synthesis, drafting, and figure/table generation from the underlying experimental data and result files described in Sections 3-5. All experimental design, IBM hardware execution, statistical analysis, and result interpretation were conducted independently of any generative AI tool; all numerical results reported in this paper are drawn directly from preserved experimental artifacts (frozen model checkpoints, IBM job records, and evaluation outputs) rather than generated or estimated by an AI tool.

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